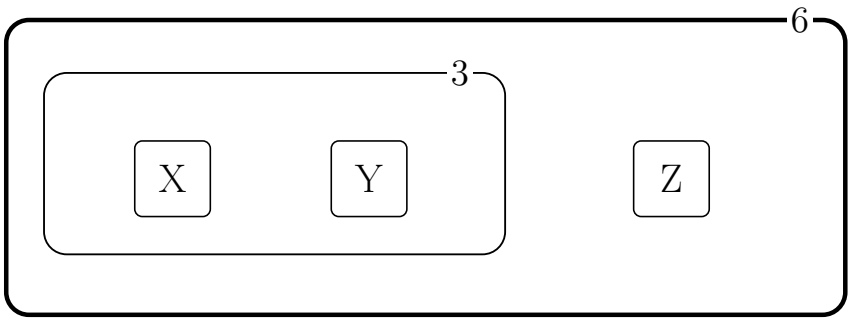
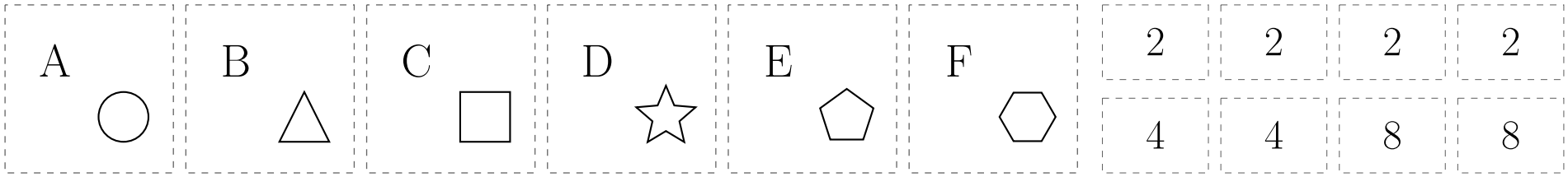
Groups can sit inside one another or be separate. They never cross. One group contains all the cards. Each card is also a one-card group. A group may contain three or more smaller groups. Every larger group has a positive label, and labels increase as you go outward. A group's size and number of cards do not determine its label.

The distance between two different cards is the label of their smallest common group. A card's distance from itself is 0. We write A–B for the distance between A and B.



X and Z ⟶ the thick group is their smallest common group ⟶ X–Z = 6.

X–Y = 3. X–X = 0.



Problem 5: Start with the clues from Problem 4. Change only A–C from 4 to 2. Can any world fit these clues?

A–C = 2

Change exactly one of the ten clues to a different value chosen from 2, 4, 8, so that a world is possible. Find every repair. For each repair, draw a world fitting all ten clues. Explain why your list is complete.

Problem 6: Choose three different cards in any allowed world. Can exactly one of their three distances be larger than the other two? Explain why your answer holds for every allowed arrangement of groups.

Problem 7: Can two balls overlap while neither contains the other? Can a ball change when you keep its radius and move its center to one of its members? Settle both questions for every allowed arrangement of groups.

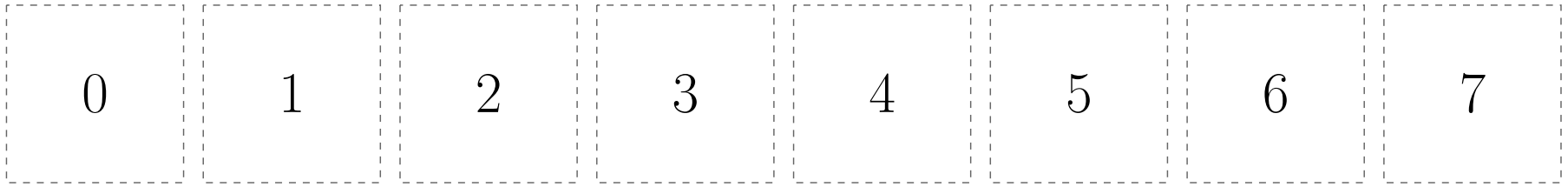
Problem 8: Use the number cards 0–7. Each card has distance 0 from itself. The distances between different cards follow these rules:

- One card is even and the other is odd: distance 8.
- Both are even or both are odd, with different remainders after division by 4: distance 4.
- They have the same remainder after division by 4: distance 2.

For cards 1 and 5, both leave remainder 1 when divided by 4. They belong to the same remainder group, so their distance is 2.

Build a world of loops with exactly these distances. Find every different ball of radius 3 and every different ball of radius 5.

Problem 9: Use the number rules to explain why the largest of the three distances between any three different number cards must occur at least twice.



Problem 10: Use cards A through F and labels 2, 4, 8. One player builds a world in secret. The other asks for distances between pairs until they can draw a world with the same distances. Try to use as few questions as possible. Trade places.